

SLIDE 10: CRIME EXPERT (Self-Introduction)

Thank you, Shivali.

While 311 calls tell us what annoys New Yorkers - the noise, the trash, the potholes - Crime Data tells us what worries them.

I'm Chintan, the Crime Expert.

As we've seen, quality of life in New York City is shaped by many factors.

My goal here is simple: To show The Millers not just where crime occurs, but what kinds of crimes are happening, and how these affect property values and their investment potential.

SLIDE 11: COMPLAIN HOTSPOTS

We started by mapping the density of all complaint types to visualize the city's stress points.

As you can see, this isn't just a scatter of dots; it's the pulse of activity, certain areas consistently show higher concentrations of crime, with some neighborhoods experiencing persistent crime issues over the years.

For The Millers, this is crucial because it highlights the neighborhoods with the most significant crime concerns.

The question isn't '**is there crime?**', but '**is it the kind of activity that affects my safety, or just my sleep?**'

SLIDE 12: BOROUGHWISE CRIME DISTRIBUTION

We break this down by offense type to find the 'flavor' of crime in each borough.

Look at Manhattan (the second column). It glows yellow for Petit Larceny. That's shoplifting, phone theft -crimes of opportunity in a retail hub.

Now, look at the Bronx (fourth column). The heatmap shifts. The intensity is higher for Assaults and related offenses.

Brooklyn shows a higher overall crime rate across multiple categories.

For the Millers, this distinction is vital: Manhattan risks are largely financial nuisances; Bronx risks are physical safety concerns.

SLIDE 13: BOROUGHWISE CRIME SEVERITY FLOW

To understand the weight of these statistics, we visualized the flow of crime severity.

On the left, we have the boroughs. In the middle, the severity: Felony – Misdemeanor – Violation.

Follow the blue path for Manhattan. You see a massive flow into Misdemeanors, which then feeds into Petit Larceny on the right supporting our earlier observation.

Now, let's contrast that with the green path for the Bronx. Here, the flow leans more heavily into Felonies and Assaults. This reinforces what we saw in the heatmap.

These insights are vital for The Millers, as they underscore how varying crime patterns directly affect neighborhood appeal and, ultimately, real estate investment potential.

SLIDE 14: BOROUGHWISE CRIME FINGERPRINTS

This is my favorite visualization: The Crime Fingerprint.

Every borough has a unique shape. As you can see, Brooklyn's crime profile is heavily skewed towards property crimes and harassment, whereas the Bronx shows a significant share of violent offenses.

We cannot price a Brooklyn home using a Bronx risk model. They are fundamentally different markets with different risk profiles.

SLIDE 15: AIR QUALITY

We've covered how crime impacts property values and now let's shift gears and turn to another critical factor - air quality.

Invisible environmental hazards silently affect both property value and personal health.

We analyzed Air Quality Index (AQI) trends to determine if the city is improving or deteriorating, and how this could influence the Millers' investment and long-term neighborhood appeal.

SLIDE 16: AIR QUALITY TRENDS ACROSS NYC (2007-2025)

The good news? The trend is our friend.

Looking at this timeline from 2007 to today, you can see a consistent downward trend in pollutants across all boroughs.

New York is cleaner today than it was a decade ago. This suggests that from a long-term environmental perspective, the Millers are buying into an improving asset.

SLIDE 17: BOROUGHWISE AIR QUALITY VARIATION

However, averages hide the local reality.

The box plot shows a sharp variation between boroughs.

Look at the Bronx (Teal box) and Manhattan (Blue box). They show higher median pollution levels and more variability compared to Queens or Staten Island.

If the Millers have respiratory concerns or young children which they do, a '**clean trend**' isn't enough-they need the '**cleanest borough**', which the data suggests is Queens or Staten Island.

I'll hand it over to my colleague, Manideep.